Lab #3: The Digital Design Process with SimUAid II

In this lab you will practice applied Boolean algebra with a 4-input Boolean expression. You will use SimUAid/LTspice and LogicAid/Karnaugh maps as tools to help you simplify your logic functions before you build them. **This lab is to be done *individually*.**

Part 1: Minterm Expansion

To get your function for this lab, **use your team number to select a unique function from the document on Canvas**. Use your team number to select the row.

For Part 1, you will take your given function and perform a minterm expansion **algebraically**. A reminder: a minterm expansion is when a Boolean function is written in standard sum of products (SOP) form, where each term in the expression is a minterm. Refer to section 4.3 in the textbook for more information.

Deliverables/Questions:

1. Write your function as a standard sum of products. Paste a picture of your work, showing your function the expansion. Please write as clearly as you can.
2. How many minterms does your function have?

Part 2: Simplifying your Boolean Function algebraically.

For this part of the lab, continue using your given function.

You will now take your team's given function and simplify it manually to a minimum SOP, while stating what theorems were used with each step of your simplification process. You can use tables 2-3 and 2-4 from the textbook as a reference. Also, peeking at a K-map to see what terms might be good to duplicate or unify is helpful.

Once you simplify your function manually, use LogicAid or K-maps to simplify your function again and compare your simplified SOP to LogicAid's or your K-map's minimum SOP. Generate a truth table of your function for Part 3.

Deliverables/Questions:

3. Paste of a picture of your work showing the theorems used for each step of the simplification process. Please write as clearly as you can.

4. Provide your fully simplified expression from LogicAid or K-maps. Is it the same?
5. Provide a truth table.
6. If the number of input combinations is determined by the function 2^n, where n is the number of inputs for a given Boolean expression, then how many combinations of inputs are needed in total for your function's logic circuit?

Part 3: Creating your Boolean function in SimUAid/LTspice

For this part of the lab, you will be using the simplified function you found in Part 2. You are to create the logic circuit in SimUAid/LTspice and implement the same techniques you learned from Lab 2, Part 3 to test the entire truth table of permutations.

Since your function has four inputs, then the binary input sequence of possible ABCD inputs is:

$$ABCD = \{0000, 0001, 0010, 0011, 0100, \ldots, 1110, 1111\}$$

You have a finite number of inputs, based on 2^n, where n is the number of inputs for a given Boolean expression.

Using the same techniques as in Lab 2**, encode the signal timing** such that each of these permutations will last 100 ns intervals. Starting with the first combination ABCD = 0000 at t = 0 ns, then the next combination ABCD = 0001 at 100 ns, and so forth until ABCD = 1111 at t = 1500 ns. This will mean that the input signals for A, B, C, and D will be changing from t = 0 ns to t = 1600 ns.

Label your inputs and outputs appropriately**.** Run the simulation. **Open a timing diagram with the appropriate scale.**

Deliverables/Questions:

7. Provide a truth table of your function using LogicAid (or by inspecting the K-map).
8. Provide the circuit schematic of your function in your simulator.
9. Provide the timing diagram of the simulation.
10. Compare the output obtained from the simulation to the theoretical values in the truth table. Does your output match that of the truth table?

Part 4: Physically Build your Circuit.

In this part of the lab, you are to build your simplified function and provide a demo showing **all input combinations and with their appropriate outputs**. Use DIP switches with pull-up/pull-down resistors to toggle the inputs easily. Use LEDs with resistors to show when the output is 1 (LED on) and 0 (LED off).You may either create a video demo, or you may come to the lab while a Digital Logic Design TA is present to demo your circuit!

Questions:

11. What pull-up/pull-down resistor values are you using and why? Please see Lab 1, Part 2.

Video demo notes:

All video demos must show you DISASSEMBLING your circuit in your video at the end of the demo. Failure to do so will result in a loss of 50% of your grade.

The demo must be no more than a 6-minute video. Add it to your Google Drive or YouTube channel and put a link to that video in your report. Make sure the sharing settings are for all with the link.

****Make sure that you put your name label on the far left of your breadboard. Make sure you state your name in the video.**

In the video, you must go over all possible combinations of your circuit. Add a truth table to this section so that we can follow your presentation.

Part 5: Summary of Deliverables

Turn in your report in either PDF or Word format, and make sure it is named: **ECE3441_StudentID#_Lab3**.

Use the following structure for your lab report:

Course Name and Number – Semester

Lab Number and Title

Name

Overview: Goals of the Lab

Briefly (~ 1 small paragraph) outline the goal of this lab, as well as the work done.

Step-by-Step Procedures and Results

This must be in chronological order of how Lab 3 should be completed. Make sure you include all of the necessary information that is mentioned in the lab and answer all questions.

Links

Include the demo link to a google drive.

Conclusion

Summarize what you have learned in this lab.